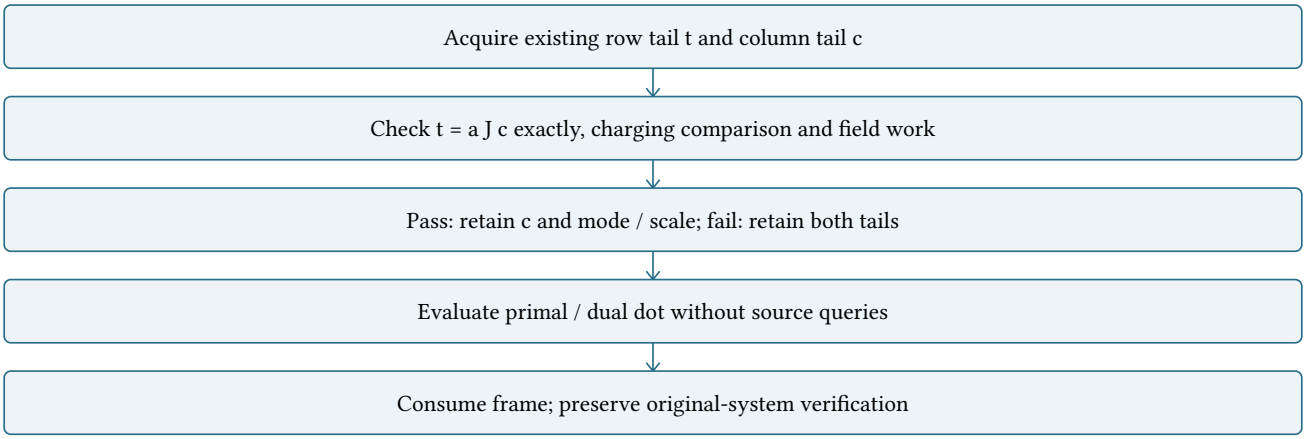
Shared border functionals and rank-one limits

2026-11-02 · Derived algebra / static source checks · Numerical trials: zero

Construction. Let q=n−1, J reverse q coordinates, and t,c be the cached row/column tails. Check t=aJc. If it holds, both lift functionals use c: $t^{\top}z=a\,c^{\top}Jz$ and $c^{\top}v$ is unchanged. If it fails, retain both original tails. No source query or inversion is introduced. Previous scalar lifts and root solution/witness checks remain unchanged.



Mode	Border slots	Primal products	Successful detection
a=0, t=0	q	0	q comparisons
a=1, t=Jc	q	q	q comparisons
general a, t=aJc	q+1	q+1	q products + comparisons
plain fallback	2q	q	charged prefix scan

All nonzero dot branches charge q additions; general scaled primal adds one product after the dot. The general mode is skipped at q=1 because no field slot is saved. Failed detection retains its overhead. Input validation, metadata, slicing and allocation are additional host work. No measured runtime or memory benefit is assigned.

Positive family. Put $C=\Sigma\, g_i\, Z_a^{\,i}$. Then $Ce_0=g$ and its first row is $(g_0,a\,g_{n-1},\ldots,a\,g_1)$. Since C commutes with Z_a, $Z_aC-CZ_\beta=(a-b)g\,e_{n-1}^{\top}$, of rank one for g≠0 and a≠b. This supplies the shared-border identity. Equality at one node need not survive Schur descent.

Information bound. Fix a=1,b=2,g_0=1. There are p^q arbitrary tails c, all with displacement rank one and nonzero first pivot. Basis queries distinguish their dot functionals. An exact self-contained encoding answering all such queries needs at least q $\log_2(p)$ bits in the worst case. L field slots and M metadata states require p^L M≥p^q. External retained data or recomputation are outside this self-contained model and must be charged.

Recurrence consequence. Order-d constant-coefficient recurrences with d initial values have at most p^(2d) encodings; they cannot cover this family when 2d is less than q. Small displacement width alone supplies no short border recurrence. This statement concerns the specified operator/query model, not a lower bound for elliptic discrete logarithms or a particular elliptic feature family.

Checks and next work. Separate frame and solver normalized-AST audits; independent coefficient checker implemented, unexecuted. No numerical, mutation, heap or performance trial. Validate all four modes, mismatch positions, q=1, both directions and counters under the full upstream cohort. Next seek a recurrence from the actual public feature map and prove its behavior under pivoting and Schur reduction, or price replay/recomputation.

Sources. DERIVATION.md, pair.py, shared_frame.py, solve_shared.py, check_pair.py and check_preparation.py. Bostan et al. (2017), equation (2) and Section 6, DOI 10.1137/16M1062855, fixes the shift/Schur conventions.